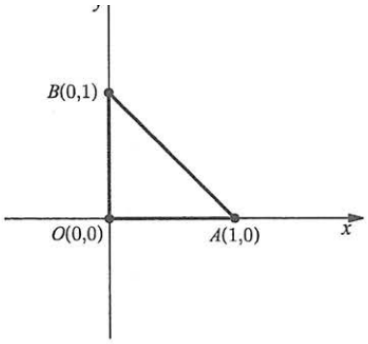
Hyperbolic Functions: Mix

Questions 1–13

Question 1. The transformation T is represented by the matrix $M = \begin{pmatrix} \cosh x & \sinh x \\ \sinh x & \cosh x \end{pmatrix}$, where $x > 0$.

a) Show that T preserves area.

b) Find the exact value of x that maps the triangle shown in the diagram below onto an equilateral triangle.



Prerequisite knowledge for Q2: Volume of Revolution, Hyperbolic Identities

Question 2. It is given that $f(x) = 4\sinh x - 6\cosh x$.

a) Show that the curve $y = f(x)$ has a turning point at $x = \frac{1}{2}\ln 5$ and determine its type.

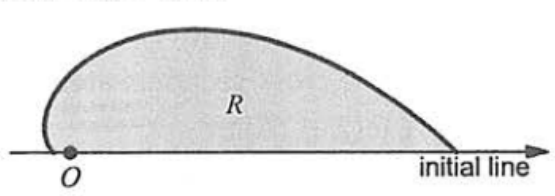
b) The arc of the curve $y = f(x)$ between $x = 0$ and $x = \frac{1}{2}\ln 5$ is rotated through 2π radians about the x-axis. Show that V, the volume generated, is given by $\pi(p + q\ln 5)$, where p and q are integers.

Prerequisite knowledge for Q3: Polar Integration

Question 3. The diagram below shows the curve with polar equation

$$r = 4\sqrt{\cosh 2\theta - \sinh 2\theta},$$

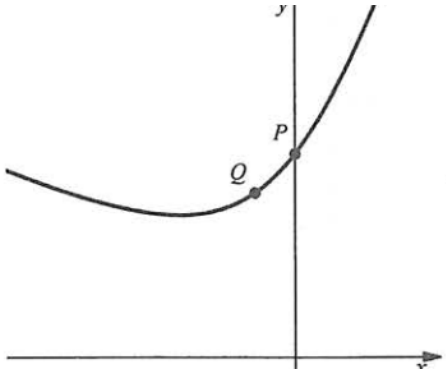
where $0 \leq \theta \leq 2\pi$.



The region R is bounded by the initial line, the curve, and the line with $\theta = \pi$. Find the exact area of R, giving your answer in the form $p + qe^{r\pi}$, where p, q and r are integers. **Prerequisite knowledge for Q4–Q5: Parametric Equations and Hyperbolic Differentiation**

Question 4. The diagram below shows the curve with parametric equations

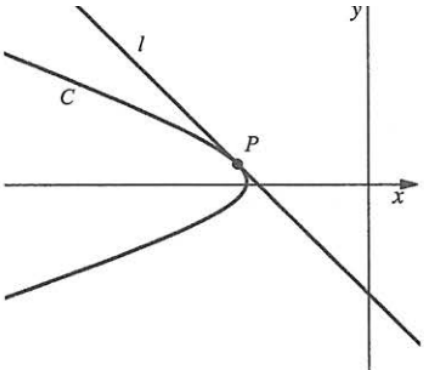
$$x = 3\sinh t - \cosh t \qquad \text{and} \qquad y = \sinh t + 2\cosh t.$$



a) The curve intersects the y-axis at P. Find the exact coordinates of the point P.
b) Q is the point on the curve when $t = 0$. Find the equation of the tangent to the curve at Q in the form $ax+by = c$, where a, b and c are integers.

Question 5. The diagram below shows the curve C with parametric equations

$$x = 5\sinh t - 8\cosh t \qquad \text{and} \qquad y = 2\cosh t - 3\sinh t.$$

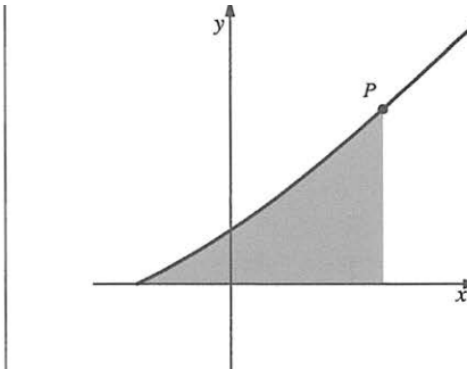


The line l is a tangent to the curve at P, where $x + y$ is at its maximum. Find the equation of the line l in the form $x + y = k$, where k is an exact constant. **Prerequisite knowledge for Q6: Parametric Equations and Hyperbolic Identities**

Question 6. The diagram below shows the curve with parametric equations

$$x = 2\sinh 2t - \cosh 2t, \qquad y = \sinh 2t$$

for $t \geq 0$. It is given that P is the point on the curve with $t = \ln 2$. The shaded region is the region enclosed by the curve, the x-axis, and the vertical line from the x-axis through P.



a) Show that the Cartesian equation of the curve is $x^2 + 3y^2 = 4xy + 1$.
b) Show that the area of the shaded region can be written in the form $a + \ln b$, where a is a rational number in its simplest form and b is a positive integer.

Prerequisite knowledge for Q7: Euler's Formula

Question 7. It is given that z is a complex number.

a) By using Euler's formula, show that $\cosh(iz) = \cos z$.

b) Assuming that the equation has complex roots, find in logarithmic form the roots of the equation $\cos z = 4$.

Prerequisite knowledge for Q8: Integration with Inverse Trigonometric Functions

Question 8.

a) Show that $3\sinh x + 5\cosh x$ can be written in the form $e^{-x}(p + qe^{2x})$, where p and q are integers.

b) Hence, using a suitable substitution, show that

$$\int_{\ln(\sqrt{3}/6)}^{\ln(1/2)} \frac{1}{3\sinh x + 5\cosh x}\, dx = \frac{\pi}{24}.$$

Prerequisite knowledge for Q9: Maclaurin Series

Question 9. It is given that

$$f(x) = \operatorname{artanh}\left(\frac{2x-1}{2-2x}\right), \qquad |x| < \frac{3}{4}.$$

a) Write $f(0)$ in the form $\ln\left(\frac{\sqrt{k}}{k}\right)$, where k is a positive integer.

b) Show that $f'(x) = \frac{2}{3-4x}$.

c) Find $f''(x)$.

d) Hence show that the Maclaurin series for $f(x)$, up to and including the x^2 term, can be written as $\ln\left(\frac{\sqrt{k}}{k}\right) + px + qx^2$, where p and q are constants to be found.

Prerequisite knowledge for Q10: De Moivre's Theorem and Mean Value

Question 10. It is given that $f(x) = \sinh ix$, where $0 \leq x \leq 2\pi$. Find the mean value of $y = (f(x))^4$.

Question 11.

a) Given that $\cosh(A+B) = \cosh A \cosh B + \sinh A \sinh B$, express $\cosh((p+1)x)$ and $\cosh((p-1)x)$ in terms of $\cosh(px)$, $\sinh(px)$, $\cosh x$, and $\sinh x$.

b) Hence show that

$$\sinh(px) = \frac{\cosh((p+1)x) - \cosh((p-1)x)}{2\sinh x}.$$

c) Hence find the sum of the series $S_n = \sinh x + \sinh 2x + \cdots + \sinh nx$ in terms of $\cosh x$, $\sinh x$, $\cosh nx$, and $\cosh((n+1)x)$.

Prerequisite knowledge for Q12–Q13: Integrating Factors for First Order Differential Equations

Question 12. Given that $y = 2$ when $x = \ln 2$, solve the differential equation

$$\frac{dy}{dx} + y\coth x = \cosh^2 x.$$

Question 13. Given that $y = 5$ when $x = 0$, solve the differential equation

$$\frac{dy}{dx} - y \tanh x = \sinh 2x.$$